

a What each validation target can establish

● Validation target	✓ Strongest claim it supports	✗ Does not establish on its own
Known motion (phantom, STRAUS, MRXCAT2.0)	representation capacity and technical error in the simulated regime	in-vivo accuracy
Paired DENSE or tagging	material-sensitive agreement under matched definitions	focal location or extent, unless measured
Scan-rescan	precision, not accuracy	bias
LGE or tissue characterisation	biological association, not motion error	pointwise displacement error
Outcome	prognostic association, not clinical utility	decision impact

b A staged programme for regional or focal claims



Pre-specify the measurement contract
estimand · reference · tolerance · data splits

Resources (Table 4)

1

Stage 1 · Known motion
vary width, extent, amplitude, timing, resolution and noise

STRAUS: 3 templates × 6 states

MRXCAT2.0: programmable generator

CMAC dynamic phantom

2

Stage 2 · Paired material reference
cine with DENSE or tagging in one session; registered; reference blinded

Stanford: 51/55 · not file-verified

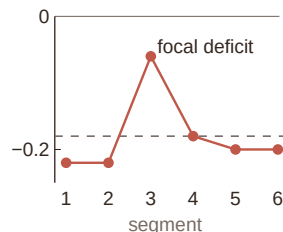
CMAC: 15 volunteers, 12 landmarks

3

Stage 3 · Tissue and decisions
LGE, responsiveness, decision impact

Clinical cohorts with LGE or outcomes

c Equal-mean test



Uniform (dashed) and focal (solid) fields share a global mean of -0.18 . Recovering the mean is not a pass.